

DATA DOMAIN

PTB

WikiText-2

TinyStories

Python

same tokenizer and
sequence length

HETEROGENEOUS TEACHER

Transformer branch
self-attention

Grassmann branch
causal pairs $Z_t, Z_{t-\Delta}$

teacher p_T

late logit fusion

$$Z_T = az^{(Tr)} + (1-a)z^{(G)}$$

STUDENT INITIALIZATION

random initialization
 θ_0

CE pretrain

CE warm-start
 θ_{CE}

CONTROLLED TRANSFER

$$\mathcal{L} = (1-\lambda)\mathcal{L}_{CE} + \lambda T^2 \text{KL}(p_T \| p_S)$$

sweep $\lambda \in [0, 0.04]$
 $T = 2; 10$ epochs

measure test PPL
initialization $\times$ domain

Research question: when does a fused teacher improve a hybrid student?